MIT 18.06SC - 2.9

Problem 22.1

$$\det\left(\begin{bmatrix} 4-\lambda & 0 \\ 1 & 2-\lambda \end{bmatrix}\right) = 0$$

$$(4-\lambda)(2-\lambda) = 0 \Rightarrow \lambda = 4 \quad \lambda = 2$$

$$\begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore$ $S = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ where the columns can be multiplied by any scalar.

$$A^{-1} = (S\Lambda S^{-1})^{-1}$$
$$= S\Lambda^{-1}S^{-1}$$

So its described by the same matrices S as A.

Problem 22.2

$$\det\left(\begin{bmatrix} 0.6-\lambda & 0.9 \\ 0.4 & 0.1-\lambda \end{bmatrix}\right) = 0$$

$$(0.6-\lambda)(0.1-\lambda) - (0.9)(0.4) = 0$$
$$(0.06 - 0.6\lambda - 0.1\lambda + \lambda^2) - 0.36 = 0$$
$$\lambda^2 - 0.7\lambda - 0.3 = 0$$

$$\lambda = 1 \qquad \lambda = -0.3$$

$$\begin{bmatrix} -0.4 & 0.9 \\ 0.4 & -0.9 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\Downarrow$$
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0.9 & 0.9 \\ 0.4 & 0.4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\Downarrow$$
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

So $S = \begin{bmatrix} 9 & 1 \\ 4 & -1 \end{bmatrix}$ and $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & -0.3 \end{bmatrix}$

As $k \to \infty$ we have $\Lambda^k \to 0$

$$S^{-1} = \frac{\begin{bmatrix} -1 & -1 \\ -4 & 9 \end{bmatrix}}{-9-4} = \frac{1}{13}\begin{bmatrix} 1 & 1 \\ 4 & -9 \end{bmatrix}$$

So $S\Lambda^k S^{-1}$:-

$$\begin{bmatrix} 9 & 1 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \left(\frac{1}{13} \times \begin{bmatrix} 1 & 1 \\ 4 & -9 \end{bmatrix} \right)$$

$$\begin{bmatrix} 9 & 0 \\ 4 & 0 \end{bmatrix} \left(\frac{1}{13} \times \begin{bmatrix} 1 & 1 \\ 4 & -9 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 9/13 & 9/13 \\ 4/13 & 4/13 \end{bmatrix}$$

In the columns of this matrix we see the steady state vector.